

Public Private Partnerships and Corporate Finance: Evidence from China and India

Abstract

In this paper, we investigate whether, Public Private Partnerships (PPPs), with readily pleadgable Government assets, reduce the underinvestment problems faced by private sector firms in emerging markets. We use PPP announcement data of partnering private sectors firms in China and India and find that: 1. PPP announcements add value to the partnering private sector firms in both India and China; 2. However, nature of firm that invests in PPP varies in both economies. Compared to Chinese firms, Indian firms face severe underinvestment problem at the time of investing in PPPs; 3. In the long run, PPPs reduce potential over-investment problem in China. Whereas, Indian firms' underinvestment problem persists; 4. Government reliance, political uncertainty and contractual features significantly affect the investment-cash flow sensitivity in both countries.

Public Private Partnerships and Corporate Finance: Evidence from China and India

1. Introduction

In the last two decades, emerging economies have experienced tremendous growth due to growing population, rapidly progressing urbanization, industrialization, and increasing integration into global supply chains. However, higher growth has led to visible infrastructure gaps that hinder such tremendous growth trajectory. It is estimated that, infrastructure demand for the 20 years to 2030 is expected to rise to USD 19.2 trillion, with Asia needing a lion's share of USD 15.8 trillion. Such huge requirement highlight the possible issues related to unmet demand for capital in the emerging economies. It is projected that US\$1.03 trillion and US\$1.025 trillion are invested to fill infrastructure gaps based on the 12th Five Year Plan in China and India respectively (Hongyan, 2010; India's Planning Commission, 2012). Given the failure of privatization program and limited capacity of both private and public sectors, Public Private Partnerships (PPPs) are gaining popularity in these markets (Engel et al., 2008).

The question we seek to answer in this study is whether Public Private Partnerships (PPPs), that have seen nine fold increase in emerging markets in the last one decade, help in reducing the capital constraints faced by private sector firms. Government involvement provides costless pledgable assets for the partnering private firm and hence enhances borrowing capacity and also reducing the cost of raising capital. This benefits cash strived private sector firms from underinvestment and helps to reduce the persistent investment-cash flow sensitivity problem of private sector firms in emerging markets. Hence, we hypothesize that PPPs benefit private sector by reducing their underinvestment problem.

It is important to note that high investment-cash flow sensitivity can also imply over-investment problem. Firms that aggressively invest can experience higher investment-cash flow sensitivity. There is every chance that, in emerging markets, cash rich private sector firms with political connections can influence and secure PPP contracts. Hence, whether private sector benefits from PPPs by reducing their underinvestment or overinvestment problem is an empirical question.

We use the PPP contracts in the two large emerging markets, namely China and India to address this question. India, for instance, has made around US 400 billion dollars worth PPP investments, which is almost half of market capitalization of the Indian capital market. However, there is virtually no research on the corporate finance side of these financial contracts. Understanding the corporate finance side of PPP contracts allows us to shed light on whether

these unique contractual agreements reduce the unmet demand for capital in emerging markets.

PPP contracts are considered as hybrid contracts as they share some features of both public and privatized- public sector firms. Similar to public sector firms, they have government backing in terms of legal and political risk, as the assets are owned by the Government. Likewise, similar to privatized public sector firms, they are managed by private sector with virtually no Government intervention. Compared to the traditional Government contracting, where construction and operation are executed as separate contracts, PPP allows bundling of contracts. In terms of risk sharing, PPP can be considered as the middle path between conventional project finance and privatization programs. The risk is shared between public and private sector unlike lying with any one of the parties as in the conventional project finance and privatization programs.

To our knowledge, ours is the first comprehensive study on PPPs in two largest emerging markets. Our study contributes towards better understanding of one of the major and fastest growing infrastructure financing contracts in the world. Our evidence provides insights on the possible benefits of these unique contractual agreements. Also, whether the intended purpose of reducing underinvestment in private sector can really be fulfilled through PPP contracts.

Using PPI database of the World Bank, we identified 138 and 124 PPP project announcements that belong to listed firms in China and India, respectively. Later, we obtained control group firms that are matched with PPP firms in terms of firm size, industry, and other firm level variables from DataStream database. Using this database, we sought answers for the following main questions. 1. Whether PPP announcements add value to the participating private sector firms? 2. Whether PPP investments reduce investment-cash flow sensitivity of partnering private sector firms in the post-PPP investment period. Also, whether such reduction is significantly lower compared to a control group of competing non-PPP private sector firms? 3. Whether higher Government participation helps to reduce capital constraints of private sector firms? 4. To what extent the variation in contract mechanism explain the cross sectional variation related to the benefits of PPP contracts? 5. Whether political uncertainty reduces the benefits of PPP contracts?

We report the following main findings: 1. Investors consider PPP announcement by partnering private firms as good news or value enhancing in both China and India. This is observed through significant positive cumulative abnormal return of 1.87% and 2.11% in China and India respectively. 2. Firms that choose to go for PPPs vary between China and India. In the case of China, firms that have low investment-cash flow sensitivity go for PPPs. Whereas in India, firms that have higher investment-cash flow sensitivity or cash-constrained firms prefer to choose PPP investments. We also find that the capital constrained firms in India go for PPP have high

growth opportunities. This implies that underinvestment drives PPP investments in India. 3. Post-PPP investment, overinvestment problem reduces among Chinese private sector firms compared to competing non-PPP firms. Whereas in India, underinvestment problems still persists in PPP firms as high growth opportunity firms continue to exhibit higher investment-cash flow sensitivity. 4. High Government equity and also projects that require more Government reliance are beneficial for partnering private sector firms in China. Whereas, Indian private sector firms benefit more from lower Government reliance. This implies that political influence and reliance matters. 5. Political uncertainty, measured in terms of election year and corresponding changes in Government, increases investment-cash flow sensitivity risk in both China and India.

Our results provide important guidance on the direction of PPPs in China and India. Our evidence can help to initiate policy debate on infrastructure gap in the emerging markets. The rest of the paper is organised in four sections. A brief introduction on the institutional features and contract mechanisms of PPPs is presented in Section two. Section three provides data description and our empirical methodology. The empirical results are presented in Section four. Section five concludes.

2. A brief introduction to PPP contracts

2.1. Definitions and characteristics of Public-Private Partnerships

There are many definitions of PPPs due to several forms of PPP contract. PPPs are generally defined as the cooperation between Government and private sector in terms of “funding, construction, renovation, management and maintenance of an infrastructure or of the provision of a service” (European Commission, 2004, p.3). The National Council for Public-Private Partnerships of the United States of America defines PPP as a “contractual arrangement between a public sector agency and a for-profit private sector developer, whereby resources and risks are shared for the purpose of delivery of a public service or the development of public infrastructure” (Akintoye et al., 2008, p.4).

In terms of PPP characteristics, Engel et al. (2010, 2008) emphasize four main characters of PPPs, including (i) bundling contracts; (ii) privately temporary ownership of assets; (iii) intertemporal risk sharing and (iv) project finance. First, similar to Privatization, bundling contracts in PPPs means that the private sector is responsible for financing, construction and operation of projects. PPPs differ from the traditional public procurement where separated contracting partners, constructors and operators, are employed to implement projects. Through

bundling, PPPs help to integrate investment expenditure and life cycle cost to avoid cost overrun and increase the efficiency of project delivery (Hart, 2003). It is even more efficient in infrastructure projects where levels of initial investment and design can determine operating cost and maintenance cost later (Engel et al., 2008). Second, temporary ownership means that the private sector usually takes over a control of assets from Government and transfers again when the contract finishes. This brings more innovation and better management because ownership is granted by the party who cares more about projects (private sector) (Engel et al., 2010; Besley and Ghatak, 2001). Third, compared to public provision where public sector bears most of risk, a risk-sharing mechanism in PPPs usually ensures project risk and life cycle cost to be transferred partially or wholly to the private sector. Finally, in terms of project financing, PPP projects are developed on the basis of project finance where cash flow generated from projects is the primary source to repay debts and dividends to sponsors (Yescombe, 2013). To ensure this economically self-contained mechanism, PPPs are implemented via the so-called Special Purpose Vehicle (SPV) (Engel et al., 2010).

The structure of PPP projects and their relationships is simulated in Figure 1. As illustrated in Figure 1, a SPV is set up, owned and run by initial sponsors (partnering private firms) and on the web of financiers, Government, customers. The private sponsors provide long-term equity capital. Government issues contract agreements, revenue sharing or guarantee and other subsidies. Sometimes, Government also injects capital to continue their interest. Under this circumstance, SPV is framed as a joint venture between private firms and Government. Financiers provide debts to the project company. The project company repays creditors directly or indirectly via Escrow agent as a third party.

2.2. Public-Private Partnerships in China and India

Public-Private Partnerships in developing countries concentrate on the BRIC emerging markets (Brazil, Russia, India and China), accounting for nearly 50% in both total value of investment and number of projects (the World Bank PPI project database, 2012). China and India are the two countries having the highest number of projects, at 18% and 12% respectively compared with 11% in Brazil and 6% in Russia. Also, PPP investment in China and India accounts for about 21% of developing countries' total value. PPPs developed as a consequence of economic reforms in China and India. They in the late 1980's and the early 1990's in China and India, when the market for public facilities was opened after long time in the planned economy (Urio, 2010). In the mean time, State-Owned Enterprises (SOEs) were contracted out and the Chinese government applied methods to encourage Foreign Direct Investment (FDI). Considerable development in

China has occurred since 1988 when the market matured and privatization massively expanded (Urio, 2010). Likewise, in India, PPPs have been strongly developed since the industrial policy opened up the economy in 1991, resulting 249 projects from 1990 to 2006 (Lakshmanan, 2008).

The emergence of PPPs is aimed to fill a huge infrastructure gap attributed to urbanization and industrialization. China and India are the second and the third largest economies in the world by GDP respectively, accounting for approximately 21.67% of total global GDP (World Bank, 2012). They are also the two biggest countries by population, which constitutes about 36.83% of total population in the world (United Nation, 2012). To adapt with their economic growth and population, urbanization, industrialization and infrastructure are essential. PPPs are widely used to deal with insufficient public investment in infrastructure, SOEs' poor supply and a change of public expenditure to soft infrastructure (Urio, 2010). PPP has been even increasingly common since Privatization in China and India has not been sufficient for infrastructure demand. This is due to the fact that both countries are pursuing a gradual approach to privatization (Gan, 2009; Kapur and Ramamurti, 2002), so there are some more delays compared with those in "big bang" privatization.

However, despite an increase in Government support for private participation in infrastructure in China and India, the enormous infrastructure gap still exists, and PPPs seem not to be encouraging enough private sector. According to Bellier (2003), less than 10 percent of funding for infrastructure came from private investors from 1990 to 2000. Likewise, in India, under the eleventh five-year plan (2007-2011), private financing accounted for about 40 percent of total investment (The High Level Committee on Financing Infrastructure, 2012). This is due to regulatory challenges and failure of previous PPP projects (Urio, 2010). In China, corruption and government credibility are still primary obstacles for Chinese firms participating in PPPs (Ke et al., 2013; Urio, 2010). Also the number of cancelled projects in China is only second to those in Argentina (the World Bank PPI project database, 2012). In fact, there are severe losses in PPP projects and even more failed projects in China, which are not reported, because the failed projects are bought back to the government and returned to conventional procurement (Urio, 2010). Meanwhile, in India, PPP projects have had to face numerous challenges on regulatory framework, risk sharing between Government and private sector, cost and time overrun (231 in 491 projects were delayed in 2008; cost overrun accounted for 51.8% in 2004 and 13.9% in 2008; the power sector experienced technical and commercial losses at 40%) (Lakshmanan, 2008).

Owing to the shortage of private participation in infrastructure, it is necessary to explore what motivates private sector to participate in PPP. Private firms really engage in PPPs when they find their sufficient incentives and a tradeoff between gains and losses of PPPs. This issue

motivates research on PPPs from private sector’s perspective.

3. Data and Methodology

3.1. Data

We develop a database by integrating several data sources. Information on PPP projects is sourced from the World Bank’s Private Participation in Infrastructure (PPI) database. Whereas, information related to financial and stock prices data of partnering private firms is obtained from Datastream. For event study analysis, the announcement dates of PPPs are sourced from the corporate announcements section of the stock exchanges’ websites. We were able to obtain 138 announcements (1988-2011) for China and 124 announcements (1996-2013) for India. For investment-cash flow sensitivity analysis, we focus on both the year of PPP announcement and also 3 or 5-year post-PPP investment period. This short term and long term analyses helps us to understand the nature of firms that engage in PPPs and the effects of PPP investment on their future investment-cash flow sensitivity relationship. We obtained 169 firm-year observations for China (1986-2012) and 215 firm-year observations for India (1991-2013). However, for long-term analysis, due to the exclusion of overlapping PPP investments, we left with 54 firm-year observations for China (1987-2014) and 63 firm-year observations for India (1997-2014).

In order to reduce the potential identification problem, we created a control group of competition non-PPP investment firms. We obtained one-to-one matched firms (for the PPP investment firms), matched by firm size and industry (based on the FTSE/Dow Jones Industrial Classification Benchmark (ICB) in Datastream). We use nearest neighbor matching method to capture the bias by matching PPP firms and Non-PPP firms by size and industry. To deal with unobservable fixed effects, δ_i, δ_t are used to capture firm fixed-effects and time-effects. The standard errors are clustered by industry to address the within-industry correlation.

3.2. Methodology

3.2.1. Public-Private Partnerships and Market Response

We use event study to evaluate the effect of PPP announcements on firm market value. The event window is 10 days before and after PPP announcements (Brown and Warner, 1985).

3.2.2. Public-Private Partnerships and Investment-Cash Flow Sensitivity

Following Fazzari et al. (1988); Hovakimian (2009), we run the following regression to estimate the differences on investment-cash flow sensitivity between PPP firms and Non PPP firms

$$\begin{aligned} \left(\frac{I}{K}\right)_{it} = & \alpha + \beta_1\left(\frac{CF}{K}\right)_{it} + \beta_2PPP + \beta_3\left(\frac{CF}{K}\right)_{it} \times PPP \\ & + f(\text{Control variables}) + \delta_i + \delta_t + \epsilon_{it} \end{aligned} \quad (1)$$

In equation (1), I is investment measured by the changes in gross fixed assets, CF is cash flow measured by income before extraordinary items, depreciation and amortization. Both are both standardized by dividing with previous years' fixed assets (K).

PPP dummy takes value 1 for PPP invested private sector firms (treatment group) and 0 for competing non-PPP invested private sector firms (control group). An interaction between *Cash flow* and PPP dummy is used to capture the differences in investment-cash flow sensitivity between the two groups.

$$PPP = \begin{cases} 1 & \text{if firms participate in PPPs (the treatment group)} \\ 0 & \text{if firms do NOT participate in PPPs (the control group)} \end{cases} \quad (2)$$

We use a list of *Control variables* that are expected to affect investment-cash flow sensitivity, namely *Firm size*, *Growth*, *Leverage*, *Age* (Hovakimian, 2009). *Firm size* is measured by the natural logarithm of total assets, and *Age* is measured by the number of years since the first year a private firm has information in Datastream. *Firm size* and *Age* are used as proxies to capture the ability to raise funds. This due to the fact that smaller firms and younger firms usually find more difficulties on raising external funds because of higher borrowing cost (Hovakimian, 2009) and adverse selection problems (Myers and Majluf, 1984). Therefore, smaller and younger firms are expected to have higher investment cash flow sensitivity. However, according to corporate life cycle hypothesis (Hovakimian, 2009), firms have more investment opportunities but less cash flow when they are young and vice versa. Hence, it is expected to have a negative relationship between cash flow and investment when firms become more mature. Another control variable is *Growth*, which is measured by the changes in sales from the previous year divided by the previous year's sales. *Growth* variable is also a proxy for growth opportunities. Likewise, *Leverage* variable, which calculated by total debt divided by total assets, can influence investment-cash-flow sensitivity in two contracting ways. Higher leverage can limit potential possibilities to raise external funds (Myers and Majluf, 1984) because of higher risk, but it also considered to be

lower financial constraints and less cash flow which helps to reduce agency problems (Hovakimian, 2009).

3.2.3. Disentangling the cause for Investment-Cash Flow Sensitivity: Under-investment or Over-investment

One potential interpretation problem associated with investment-cash flow sensitivity is that it can be attributed to both under and over investment problem. Investment-cash flow sensitivity is supported by Myers and Majluf (1984) who claim that firms with high information asymmetries cannot be financed by external sources since they are too expensive and unavailable. As a result, managers may forgo positive NPV project and suffer from underinvestment. Investment-cash flow sensitivity also explained by Jensen (1986) in terms of agency problems. Accordingly, owing to the conflict between managers and shareholders, managers use the free cash flow to pursuit negative NPV (overinvest) at their own best interest and their shareholders expense, which results in higher dependence of investment on internal cash flows.

We address this potential issue by exploring whether investment-cash flow sensitivity is caused by underinvestment or overinvestment. Firms with high investment opportunities (high Tobin's q) may suffer more information asymmetries and have less pledged assets, resulting in high dependence on internal cash flow (Myers and Majluf, 1984; Pawlina and Renneboog, 2005). Meanwhile, according to Jensen and Meckling (1976), managers tend to overinvest free cash flow to pursuit their pecuniary and non-pecuniary benefits of larger firm size. Firms with low growth opportunities (low Tobin's q) have more overinvestment problems due to the shortage of positive NPV projects, which lead to high investment-cash flow sensitivity.

We use Tobin's q (*tobin q*) with one year lag to capture investment opportunities and then follow Dawson and Richter (2006) to interpret 3-way interaction $ppp * tobin_{q_{t-1}} * cash\ flow$. The approach we conduct is first to run the below regression with 3-way interaction.

$$\begin{aligned} \left(\frac{I}{K}\right)_{it} = & \alpha + \beta_1\left(\frac{CF}{K}\right)_{it} + \beta_2PPP + \beta_3tobin_{q_{t-1}} + \beta_4\left(\frac{CF}{K}\right)_{it} \times PPP + \beta_5\left(\frac{CF}{K}\right)_{it} \times tobin_{q_{t-1}} \\ & + \beta_6PPP \times tobin_{q_{t-1}} + \beta_7PPP \times tobin_{q_{t-1}} \times \left(\frac{CF}{K}\right)_{it} + f(Control\ variables) + \delta_i + \delta_t + \epsilon_{it} \end{aligned} \quad (3)$$

Subsequently, we compute simple slopes of *Investment* variable on *Cash flow* variable where the moderator variables, *ppp* and *tobin q_{t-1}*, are held constant at different combinations of high and low values. We then compare those simple slopes and test whether their differences are

significant from zero in predicting the *Investment* variable. Consequently, there are six pairs of slopes.

- (1)(PPP & high tobin q) - (PPP & low tobin q)
- (2)(PPP & high tobin q) - (NonPPP & high tobin q)
- (3) (PPP & low tobin q) - (NonPPP & low tobin q)
- (4) (NonPPP & high tobin q) - (NonPPP & low tobin q)
- (5) (PPP & high tobin q) - (NonPPP & low tobin q)
- (6) (PPP & low tobinq) - (NonPPP & high tobinq)

In developing hypothesis on whether the differences in investment-cash flow sensitivity in PPP firms is caused by underinvestment or overinvestment, we consider the first 3 pairs (Pair 1, Pair 2 and Pair 3). This is due to the fact that, with two variables (*PPP* and *tobin q*), we need to keep one moderator constant, the other one is changed from high to low values. Pair 4 follows the rule, however, not included in our analysis as it is related only to the changes in investment-cash flow sensitivity of Non-PPP firms. Accordingly, the hypothesis development is illustrated in Table I.

In Table I, the (1,1) entry, that refers to Pair 1, captures simple slope when we keep the moderator *PPP* constant and let the moderator *tobinq* changes from high value to low value (one standard deviation above and below the mean respectively)(Dawson and Richter, 2006). The (1,1) entry value is positive and statistically significant means that PPP firm experiences the higher investment-cash flow sensitivity when the investment opportunities (*tobinq*) is higher. In other words, such firms face underinvestment problem as they are capital constrained even with high growth investment opportunities. On the other hand, if the(1,1) entry is negative then it indicates that such firms face over investment problem low growth opportunity firms enjoy lower capital constraints (Vogt, 1997).

Likewise, the (2,1) cell, regarding Pair 2, illustrate the difference of simple slopes when the moderator *PPP* dummy variable changes from high value 1 (referring to PPP) to low value 0 (referring to Non-PPP) and the moderator *tobin q* remains high level (one standard deviation above the mean)(Dawson and Richter, 2006) in both cases. A positive difference indicates high q PPP firms having higher sensitivity of cash flows to investments compared to non-PPP firms. This supports underinvestment hypothesis (Vogt, 1994; Pawlina and Renneboog, 2005).

In the (3,1) cell and corresponding Pair 3, the moderator *PPP* dummy changes from high value 1 to low value 0 and the moderator *tobin q* is at low level (one standard deviation below the mean)(Dawson and Richter, 2006)) in both cases. In this case, a higher sensitivity implies over-investment problem as low q PPP firms display higher demand for investment.

For robustness we use Bonferroni correction to reduce the probability of type I errors. Bonferroni correction is conducted by calculating adjusted-p value. Adjusted-p value is measured by dividing the critical p-value by the number of simultaneous tests (Miller, 1966; Dawson and Richter, 2006).

3.2.4. The Effect of The Public Sector on PPP Firms' Investment-Cash flow Sensitivity

To indicate the effects of the public sector on PPP firms, we run the following regression

$$\begin{aligned} \left(\frac{I}{K}\right)_{it} = & \alpha + \beta_1\left(\frac{CF}{K}\right)_{it} + \beta_2 Public\ variable + \beta_3\left(\frac{CF}{K}\right)_{it} \times Public\ variable + f(Control\ variables) \\ & + \delta_i + \delta_t + \epsilon_{it} \end{aligned} \quad (4)$$

In the equation (4), the *Public variable* can be categorized by *Public Equity* that is measured by the ratio of public investment to PPP infrastructure total investment to capture government involvement in PPP firms. This is achieved in the World Bank's PPI database by subtracting the proportion of private investment from 1.

With the purpose of clarifying actual effects of the public sector on PPP firms, we again follow Dawson and Richter (2006) to interpret 3-way interaction *public equity(PE)*tobin q_{t-1}*cash flow(CF)*. The approach we conduct is first to run the below regression with 3-way interaction.

$$\begin{aligned} \left(\frac{I}{K}\right)_{it} = & \alpha + \beta_1\left(\frac{CF}{K}\right)_{it} + \beta_2 tobin\ q_{t-1} + \beta_3 PE + \beta_4\left(\frac{CF}{K}\right)_{it} \times tobin\ q_{t-1} + \beta_5\left(\frac{CF}{K}\right)_{it} \times PE \\ & + \beta_6 tobin\ q_{t-1} \times PE + \beta_7 tobin\ q_{t-1} \times PE \times \left(\frac{CF}{K}\right)_{it} + f(Control\ variables) + \delta_i + \delta_t + \epsilon_{it} \end{aligned} \quad (5)$$

Subsequently, we compute simple slopes of *Investment* variable on *Cash flow* variable when the moderator variables, *Public Equity (PE)* and *tobin q*, are held in constant at different combination of high and low values. We then compare those simple slopes and test whether their differences are significant from zero in predicting the *Investment* variable. We interpret these results and link them with underinvestment and overinvestment problems like the previous section 3.2.3.

3.2.5. The Effect of Contract Mechanism on Investment-Cash flow Sensitivity

Contract mechanism play a significant role on the success of PPP ventures. The contracts can be mainly classified based on the awarding method, nature of the project and the method

of sharing revenue. PPP can be awarded either through competitive bidding process or direct negotiations with the Government. Based of the nature of project, PPPs can be broadly classified as Greenfield and Concession projects. PPP venture future revenues can be shared either through fixed payments from the government or from revenue sharing through user fee from the customers.

Projects that are awarded through competitive bidding, green field projects and revenue generated through user fees are more independent as they are less reliant on the Government. On the other hand, Government connections and relationship would benefit if the projects are awarded through negotiations, receive government concessions and also future revenue is determined by the Government. It is not clear whether less or more reliance on Government is beneficial for the private sector firms. A higher reliance on Government reduces the risk of future cash flows as firms can have more reliable estimates on their future cash flows. On the other hand, higher reliance carries higher uncertainty due to political risk. Due to long term nature of the projects, change in Government or Government officials can have a negative impact on the future cash flows related to the project.

India being a democratic country, firms would benefit more from less Government reliant projects compared to China. On the other hand, higher dominance of the Chinese Government in business implies that Chinese firms would benefit more with higher Government dependency.

$$\begin{aligned} \left(\frac{I}{K}\right)_{it} = & \alpha + \beta_1\left(\frac{CF}{K}\right)_{it} + \beta_2 Project\ dummy + \beta_3\left(\frac{CF}{K}\right)_{it} \times Project\ dummy + f(Control\ variables) \\ & + \delta_i + \delta_t + \epsilon_{it} \end{aligned} \quad (6)$$

In the equation (6) , the *Project dummy* can be categorized by types of PPPs, awarding methods or the sources of revenue.

Regarding to Types of PPPs

$$Project\ dummy = \begin{cases} 0 & \text{if Concession} \\ 1 & \text{if Greenfields} \end{cases} \quad (7)$$

Concerning Awarding Methods

$$Project\ dummy = \begin{cases} 0 & \text{if Competitive bidding} \\ 1 & \text{if Direct negotiation} \end{cases} \quad (8)$$

With regards to Sources of Revenue

$$Project\ dummy = \begin{cases} 0 & \text{if Fixed payments} \\ 1 & \text{if User fees} \end{cases} \quad (9)$$

A major obstacle to using fixed-effects model is that it will sweep out the effects of time-invariant variables such as project dummy variables. In these cases, according to Chatelain and Ralf (2010), pre-test estimator will be applied to choose a suitable estimation among Mundlak-Krishnakumar, Hausman Taylor and restricted GLS. The pre-test, which is based on the case of Mundlak-Krishnakumar, is estimated

$$y_{it} = \beta_1(X_{1it}) + \beta_2(X_{2it}) + \beta_3(X_{3it}) + \pi_1\bar{X}_{1i} + \pi_2\bar{X}_{2i} + \pi_3\bar{X}_{3i} + \gamma Z_i + \alpha_i + \epsilon_{it} \quad (10)$$

In the equation (10), X is time-varying variable, Z is time-invariant variable, $\bar{X}$ is the average over time of each time-varying variable. According to Chatelain and Ralf (2010), based on the estimation of π_1, π_2, π_3 , the suitable estimation will be chosen.

If π_1, π_2, π_3 are different from zero, the Mundlak-Krishnakumar estimation is chosen. In fact, it is unrestricted GLS of equation (10) but still includes time-invariant variables in the model (Mundlak, 1978).

If π_1, π_2, π_3 are not different from zero, the restricted GLS is chosen.

Otherwise, Hausman Taylor estimation is chosen. For example, if π_1 is not different from zero and π_2, π_3 is different from zero, X_1 is an exogenous variable, X_2 and X_3 are endogenous variables. X_1 is served as an internal instrumental variable to identify endogenous time-invariant variable (Hausman and Taylor, 1981). In my analysis, project dummies are considered as endogenous time-invariant variables because the changes in firms' investment can assumed to influence the way firms choose contract mechanisms. Also, Hausman Taylor estimator in Stata, as indicated by the command `xthtaylor`, can not allow to cluster industry to capture industry correlation. Therefore, my study includes Industry variable as a set of dummies variable as well as exogenous time-invariant variables in the regression to capture industry effects.

4. Results

4.1. Public-Private Partnerships and Market Response

Table II presents event study based results that are aimed to evaluate market response on PPP investment announcements by listed private sector firms in China and India. The results confirm that, on average, announcement of Public-Private Partnerships has significant positive impact on firm value. Both Chinese and Indian PPP firms experience statistically significant positive average abnormal returns (AAR) on PPP announcement day (Day 0). This is consistent with the Shareholder Value Maximization hypothesis and the traditional valuation theory which associates strategic investment decisions to possible increase in future cash flows (Woolridge and Snow, 1990). Figure 2 depicts the trend of CAR over the event window from $t=-10$ to $t=10$. Overall, Chinese and Indian firms' witness significant increase in their wealth. However, there appears to be leakage of information in China as the Chinese firms react much early to the public announcement date.

4.2. Public-Private Partnerships and Investment-Cash flow sensitivity

We report whether PPP firms are significantly different, in terms of firm characteristics, compared to competing non-PPP invested private firms in Table III and Table IV. The main objective of this mean difference analysis is to understand how the treatment and control group firms vary at the time of PPP and in the post-PPP investment period (3 years).

As shown in the tables, both PPP firms in China and India are slightly larger than competing non-PPP invested firms. Chinese PPP firms are valued better than their counterparts who do not invest in PPPs. Such difference does not exist in India. Also, Chinese PPP firms are older than non-PPP invested firms. On the contrary in India, PPP firms are younger. It is important to note that, Chinese PPP firms, due to lower leverage ratio, are less burdened compared to non-PPP counterparts. However, the trend is exactly the opposite in India. Indian PPP firms are more debt burdened. These results highlight the nature of firm that engages in PPP in both economies. In China, older, matured, better valued and high cashflow firms engage in PPPs. On the other hand, in India, firms that are younger with relatively higher debt burden engage in PPPs. This indicates that, on a relative basis, Indian firms that invest in PPPs might consider PPPs as a method of reducing their underinvestment problem.

Table V and Table VI compare investment-cash flow sensitivity between PPP firms and non-PPP firms in PPP year and 3-year post-PPP. The 1st and 3rd columns report the estimation

results for Equation 1. The objective of reporting results is to understand firms characteristics and their relative sensitivity of investments to their corresponding internal cash flows.

At the time of engaging in PPP contracts, as can be seen in Column 1 of Table V, overall, Chinese firms suffer from investment-cash-flow sensitivity problem. This is evident from the positive and significant coefficient for *cash flow* variable. However, those firms that engage in PPP investments has relatively lower investment-cash-flow sensitivity. This is evident from the negative and significant coefficient for the interaction variable between *ppp dummy* and *cash flow*. This is inline with Table III, descriptive statistics that less debt burden Chinese firms engage in PPP compared to competing firms of same size in the same industry.

In contrast, the results for Indian firms, as can be seen in Column 3 of Table V, indicate that PPP investment firms have higher investment-cash flow sensitivity relationship compared to non-PPP investment firms. This result is also inline with Table III, indicating that Indian PPP firms are relatively more cash burdened at the time of going for PPP investment.

The results for post-PPP investment, as indicated in the 1st and 3rd columns of Table VI, show that Chinese PPP firms still maintain lower investment-cash flow sensitivity than non-PPP firms after 3 years engaging in PPP contracts. Whereas, Indian firms continue exhibit higher investment-cash flow sensitivity relationship compared to non-PPP investment firms.

Now we turn our analysis on whether over or under investment drives higher cash flow sensitivity in China and India.

4.2.1. Underinvestment or Over-investment

In order to explore reasons for different investment-cash flow sensitivity in China and India, we use 3-way interaction $ppp * tobin\ q_{t-1} * cash\ flow$ and interpret it by comparing simple slopes. The 2nd and 4th columns of Table V and Table VI report the estimation results for 3-way interaction in Equation 3. We then apply the slope difference method in Table VII and Table VIII.

During the year of PPP investment(PPP year), as shown in Table VII, there are no significant differences in the slopes of Chinese firms while Indian PPP firms witness significant positive slope differences between PPP firms with high *tobin q* and non-PPP firms with high *tobin q*. This indicates that PPP firms with high *tobin q* experience higher investment-cash-flow sensitivity than non-PPP firms with high *tobin q*. The slope differences between PPP firms with low *tobin q* and non-PPP firms with low *tobin q* are not statistically significant. Hence, the higher investment-cash flow sensitivity of PPP firms compared to non-PPP firms is found in the sub-group of firms with higher *tobinq* or firms with higher growth opportunities. This result supports underinvestment hypothesis (Vogt, 1994). This indicates that Indian firms depend much on

internal cash flow to circumvent the adverse selection problem (Vogt, 1994; Myers and Majluf, 1984; Stiglitz and Weiss, 1981).

We now turn to post-PPP sensitivity analysis. This analysis aims to understand the effect of PPP investment on the investment cash-flow sensitivity of the invested firms. Table VIII reports results. As indicated in Panel A and Hypothesis Matrix, for Chinese firms, we find significant negative slope differences between PPP firms with low *tobin q* and non-PPP firms with low *tobin q*. The slope difference between PPP firms with high *tobin q* and non-PPP firms with high *tobin q* is not significant. Therefore, in China, the lower investment-cash flow sensitivity of PPP firms than non-PPP firms only occurs to the sub-group of firms with lower *tobin q* or lower growth opportunities. This result supports over-investment hypothesis. Managers may use free cash flow to over-invest in negative NPV projects (Jensen, 1986). In China, partnering private firms may reduce overinvestment problems through PPP mechanism. Panel B and Hypothesis Matrix reports Indian PPP firms' results. The results indicate persistent higher investment-cash flow sensitivity for high *tobin q* cases.

4.2.2. The Role of Government Commitment and Political Uncertainty

PPP projects being Government involved, Government commitment and political uncertainty can have a significant impact on the cash flow position of the participating private firms. We test this conjecture, by using equity participation by the Government as a measure of Government commitment or involvement in a given PPP venture. We use election year dummy as our measure of political risk. We conjecture that PPPs that enter contracts in the election year would face higher uncertainty as possible policy changes of the new Government can impact the cash flows associated with the PPP investments.

Table IX report the results of regression on Equation 4 for five years post the contract period. We use five year period keeping the election cycle in perspective. We also run three year period based regression that are not reported in the paper. We find that, for Chinese PPP firms, higher equity contribution from the Government reduces investment-cash flow sensitivity of the partnering private sector firm in post-PPP period. However, in the case of India, it is exactly the opposite: higher equity contribution from the Government increase participating private sector firms' investment-cash-flow sensitivity. This indicates that higher commitment can have mixed results. However, in the case of political uncertainty, measured by election year dummy, the results are similar for both China and India. Those PPP projects that are awarded during the election year face higher investment-cash flow sensitivity.

Table X clarifies reasons for the influence of Government involvement on firms' sensitivity. As

can be seen on Panel A and Hypothesis matrix of Table X, there is a significant negative slope difference between firms with high *public equity*, low *tobin q* and firms with low *public equity*, low *tobin q*. This indicates that firms with more Government investment have less overinvestment or Government involvement helps Chinese firms to reduce overinvestment problems. The results for Indian firms, as indicated in Panel B, are insignificant.

4.2.3. The Role of Contractual Agreements on Investment-Cash-Flow Sensitivity

Our final analysis explores the role of contractual agreements on investment-cash-flow sensitivity relationship, as shown on Table XI. The contractual methods are mainly classified under three groups. The first classification is based on whether the project is a new green field project with full autonomy of the private partner or a concession project, where private partner takes over the management of existing public sector firm. We find that green field projects experience higher investment-cash flow sensitivity in China whereas it is opposite in India. This is consistent with the level of Government equity participation result in the previous section. It is suggested that Chinese private sector firms benefit with higher reliance on Government whereas Indian private sector firms benefit more from independent projects with total autonomy.

The second classification is based on the awarding method of the PPP contract. As discussed earlier, PPPs can be awarded either through competitive auctions or through direct negotiations with the Government. The results indicates that the awarding method matters in China. Chinese private sector firms benefit with lower investment-cash-flow sensitivity when the contracts are awarded through direct negotiations. However, the awarding method is not significant in India.

The final classification is based on the source of revenue for the private partner. The private partner in a PPP project can collect revenue directly by charging fee from the customers or receive a fixed payment from the Government. The higher uncertainty for partnering private firms associated with direct collection from the future users of the project is experienced both in China and India. In both countries, contracts that collect future revenue through user fee experience higher investment-cash-flow sensitivity during the post-PPP investment period.

In summary, one interesting finding from the contracting method analysis is that Chinese firms benefit more from Government intervention, whereas, Indian firms benefit from less Government intervention.

5. Conclusion

Public-Private Partnerships (PPPs) are gaining significance as a major investment partnership between public and private sector, aimed at addressing the growing infrastructure gap in emerging markets. PPPs come with the benefit of readily pleadable Government assets that can help private sector to invest in large infrastructure projects which otherwise increase their investment-cash flow sensitivity.

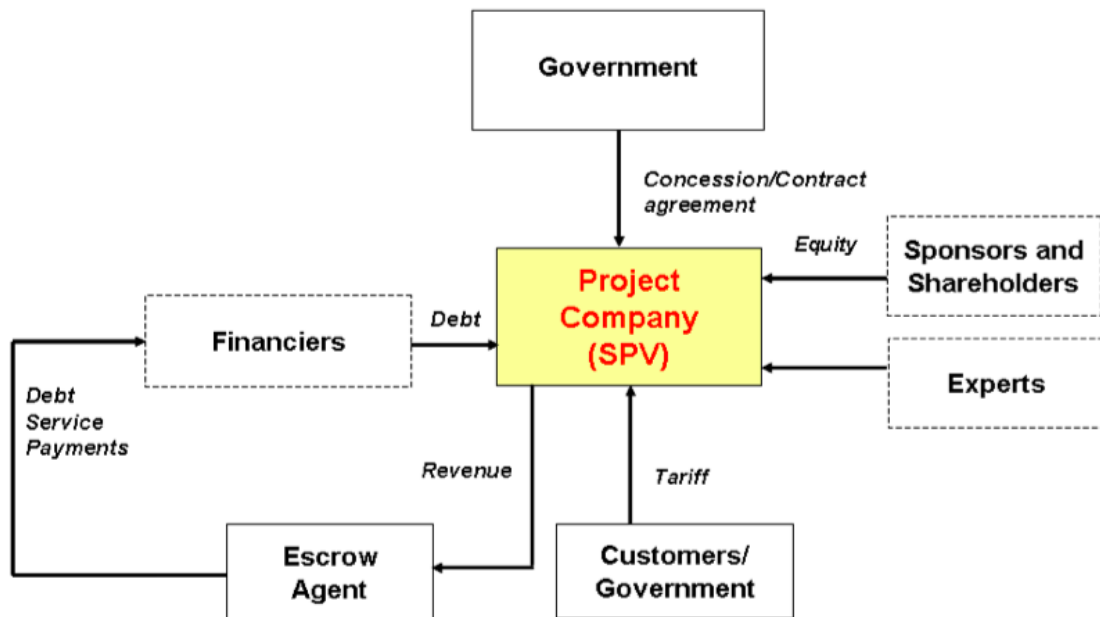
We test whether this inherent benefit really reduce underinvestment problem that can be evident from reduction in the investment-cash-flow sensitivity of private sector firms. We use the largest emerging markets, namely, Indian and China to test this conjecture. We also try to attribute whether change in investment-cash-flow sensitivity is driven by under or over investment problem. This is important as any reduction in investment-cash flow sensitivity does not guarantee reduction in underinvestment problem. If the reduction is experienced mainly by a sub-group of firms that have low growth opportunities then it reflects potential over-investment problem.

Our results indicate that Chinese private sector firms, that have less dependency on their cash flows for investment, invest in PPPs. Their investment-cash flow sensitivity further reduces in the post-PPP investment period. However, the reduction is mainly for a subgroup of firms that have low growth opportunities. This indicates that Chinese private sector firms may reduce overinvest problems through PPP investments. We further support this argument by exploring possible dependency of Chinese private firms on Government. We find that higher dependency on the Government, in the form of Government investment and contractual arrangement benefits Chinese private sector firms through reduction in their investment-cash flow sensitivity. However, such benefits are mainly exploited by private sector firms that have lower growth opportunities. This means that high Government intervention may circumvent the agency conflict between managers and shareholders, which in turn reduce overinvestment problems.

In the case of India, private sector firms with high sensitivity of investment to their cash flows invest in PPPs. We find that, unlike Chinese private sector firms, Indian private sector firms, that investment in PPPs, mainly suffer underinvestment problem. Contrary to the Chinese firms, higher reliance on Government is detrimental to Indian firms.

In summary, our research highlights the problems associated with PPP contracts. We find that the benefits of PPP arrangements are sensitive to various institutional and contractual arrangements.

Figure 1: Contractual Relationships in a Special Purpose Vehicle (SPV)



Source: UNESCAP (2011)

Figure 2: China & India - CAR graph

This figure presents graphs of Cumulative abnormal return(CAR) with the event window from $t=-10$ to $t=10$ of 138 PPP announcements in China and 124 announcements in India. The figure for CAR based on Table II

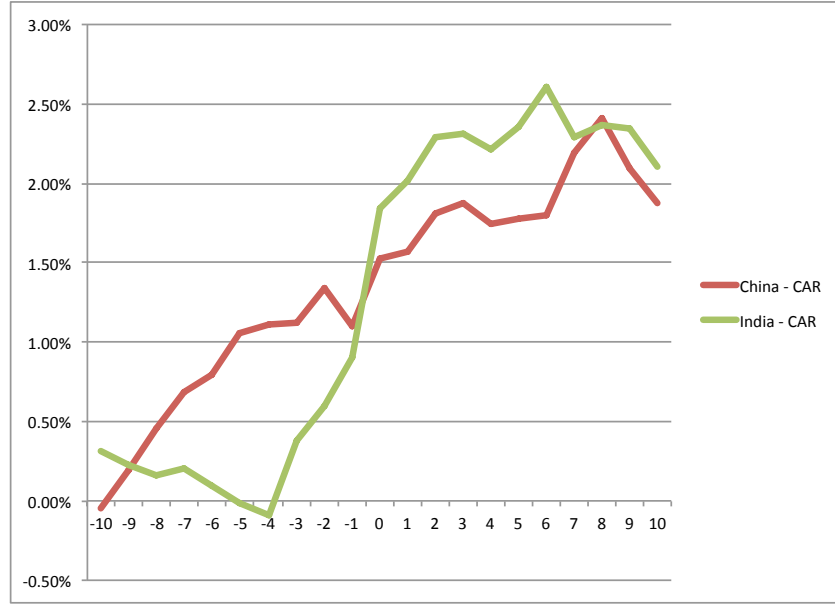


Table I: Hypothesis Development Matrix on Causes of Investment-Cash flow sensitivity

This table indicates the 3x2 matrix for hypothesis development. Each cell (or entry) shows the value of changes in simple slopes when combining the moderator PPP and the moderator tobin q. We consider 2 different values in each cell to interpret, including positively statistically significant and negatively statistically significant.

	PPP - PPP	PPP - Non-PPP
High q - Low q	(+): PPP has underinvestment (-): PPP has overinvestment	No prediction
High q - High q	No prediction	(+): PPP has more underinvestment than Non-PPP (-): PPP has less underinvestment than Non-PPP
Low q - Low q	No prediction	(+): PPP has more overinvestment than Non-PPP (-): PPP has less overinvestment than Non-PPP

Table II: China & India - Event study

The table presents an event study to evaluate the effects of 138 PPP announcements of Chinese firms and 124 PPP announcements of Indian firms on firm market value. The event window is 10 days before and after PPP announcements. An abnormal return is calculated by adjusting daily return by expected return, which is approximated by the market model with 250 day estimate window. In Panel A, an average abnormal return (AAR) is measured by averaging the abnormal return over PPP announcements on the same day in the event window. A cumulative abnormal return (CAR) is calculated over 20 days from $t=-10$ to $t=10$. In Panel B, A cumulative abnormal return (CAR) is calculated over different windows, including -10 to +10, -1 to +1, 0 to +5, 0 to +1. t-test is used to test whether the average abnormal return is significant different from zero or not. ***, **, * indicate significant at 1%, 5% and 10% level

Panel A		China			India		
Day relative to event	AAR	T-stat	CAR	AAR	T-stat	CAR	
-10	-0.05%	-0.2449	-0.05%	0.32%	1.3216145	0.32%	
-9	0.24%	1.1449	0.19%	-0.08%	-0.34982	0.23%	
-8	0.27%	1.2578	0.46%	-0.08%	-0.316192	0.16%	
-7	0.23%	1.0759	0.69%	0.04%	0.1795917	0.20%	
-6	0.11%	0.5055	0.80%	-0.11%	-0.444861	0.09%	
-5	0.26%	1.2096	1.05%	-0.11%	-0.448888	-0.01%	
-4	0.06%	0.2741	1.11%	-0.08%	-0.327684	-0.09%	
-3	0.01%	0.0394	1.12%	0.47%	1.9581724	0.37%	
-2	0.22%	1.0176	1.34%	0.22%	0.9166105	0.59%	
-1	-0.24%	-1.1368	1.10%	0.32%	1.3224278	0.91%	
0	0.43%	2.0233	1.53%	0.94%	3.9387457	1.85%	
1	0.04%	0.1863	1.57%	0.18%	0.7349251	2.02%	
2	0.25%	1.1624	1.82%	0.27%	1.1214531	2.29%	
3	0.06%	0.2617	1.87%	0.03%	0.1050315	2.31%	
4	-0.13%	-0.5979	1.74%	-0.10%	-0.429246	2.21%	
5	0.03%	0.1316	1.77%	0.15%	0.6110889	2.36%	
6	0.03%	0.1417	1.80%	0.25%	1.0298843	2.60%	
7	0.40%	1.8539	2.20%	-0.31%	-1.306845	2.29%	
8	0.21%	1.0036	2.41%	0.07%	0.3014681	2.36%	
9	-0.32%	-1.4877	2.09%	-0.02%	-0.082297	2.34%	
10	-0.22%	-1.0473	1.87%	-0.24%	-0.989551	2.11%	
Panel B		China		India		No of announcements	
Windows	CAR	T-stat	CAR	T-stat	China	India	
(-10,+10)	1.87%	8.7373***	2.11%	8.8456***	138	124	
(0,+5)	0.68%	3.1674***	1.45%	6.0820***			
(0,+1)	0.47%	2.2096**	1.11%	4.6737***			
(-1,+1)	0.23%	1.0810	1.43%	5.9961***			

Table III: Descriptive Analysis - PPP & Non-PPP firms in the PPP year

This table provides the mean firm-level characteristics of the sample firms, the difference of means between PPP firms and non-PPP firms along with t-test. The mean value is reported in the years when firms have PPP projects. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Firm size is measured by the natural logarithm of total assets. Growth presents the changes in sales from the previous year divided by the previous year's sales. Leverage is calculated by total debt divided by total assets. Age is measured from the year of firms incorporation. Tobin q_{t-1} (with one year lag) to capture investment opportunities. ***, **, * indicate significant at 1%, 5% and 10% level.

Country	Variable	PPP	Non-PPP	Difference	t test	N	
Panel A						PPP	Non-PPP
China	Size	7.062083	6.86678	0.195303	1.9695**	168	148
	Growth	0.4922446	0.4177145	0.0745301	0.3971	160	140
	Leverage	0.2651459	0.3022212	-0.0370753	(-1.6565*)	168	148
	Investment	0.5442851	1.809764	-1.2654789	-1.1051	159	140
	Age	12.48521	8.689189	3.796021	4.5937***	169	148
	<i>Tobin q_{t-1}</i>	2.212313	1.579672	0.632641	2.4099**	147	122
	Cash flow	0.6594399	5.677211	-5.0177711	-0.9978	159	140
Panel B						PPP	Non-PPP
India	Size	7.665063	7.537576	0.127487	1.7625*	213	149
	Growth	0.6523115	0.30125	0.3510615	1.1125	186	145
	Leverage	0.3996854	0.3114903	0.0881951	4.6258***	213	149
	Investment	1.42221	0.2914012	1.1308088	1.5839	194	145
	Age	7.772093	9.38255	-1.610457	(-2.9751***)	215	149
	<i>Tobin q_{t-1}</i>	2.511143	2.063233	0.44791	0.7724	175	133
	Cashflow	0.881392	0.9963403	-0.1149483	-0.2918	192	144

Table IV: Descriptive Analysis - PPP & Non-PPP firms in 3-year Post-PPP

This table provides the mean-firm level characteristics of the sample firms, the difference of means between PPP firms and non-PPP firms along with t-test. The mean value is reported in the 3 years after the PPP year. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Firm size is measured by the natural logarithm of total assets. Growth presents the changes in sales from the previous year divided by the previous year's sales. Leverage is calculated by total debt divided by total assets. Age is measured from the year of firms incorporation. $tobin\ q_{t-1}$ (with one year lag) to capture investment opportunities. ***, **, * indicate significant at 1%, 5% and 10% level.

Country	Variable	PPP	Non-PPP	Difference	t test	N	
Panel A						PPP	Non-PPP
China	Size	7.317187	7.064299	0.252888	1.6726*	52	48
	Growth	0.4819176	0.2116215	0.2702961	0.7691	52	48
	Leverage	0.2355461	0.2789306	-0.0433845	-1.453	52	48
	Investment	0.448708	0.4438703	0.0048377	0.0185	52	48
	Age	13.27778	10.27778	3	2.6301***	54	54
	$tobin\ q_{t-1}$	2.049608	1.697451	0.352157	0.6185	51	51
	Cash flow	0.3284709	0.4250168	-0.0965459	-0.9132	52	48
Panel B						PPP	Non-PPP
India	Size	7.872646	7.732555	0.140091	1.1515	62	60
	Growth	0.2992297	0.1469691	0.1522606	1.8279*	62	60
	Leverage	0.3979612	0.3170071	0.0809541	2.5584**	62	60
	Investment	0.5396645	0.2965039	0.2431606	1.5361	62	60
	Age	9.761905	10.14286	-0.380955	-0.4847	63	63
	$tobin\ q_{t-1}$	1.622787	1.698413	-0.075626	-0.2165	61	63
	Cash flow	0.6333124	0.9538211	-0.3205087	-1.1668	62	60

Table V: China & India: Investment-Cash Flow Sensitivity - PPP firms & Non-PPP firms in PPP years

This table compares investment-cash flow sensitivity between PPP firms and Non-PPP firms. Regression (1) & (2) for Chinese firms, regression (3) & (4) for Indian firms. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Size is measured by the natural logarithm of total assets. Growth presents the changes in sales from the previous year divided by the previous year's sales. Leverage is calculated by total debt divided by total assets. Age is measured from the year of firms incorporation. PPP is a dummy variable which takes value 1 for firms invested in PPP projects, else zero for firms which are matched by industry and firm size. Tobin q_{t-1} (with one year lag) to capture investment opportunities ***, **, * indicate significant at 1%, 5%, and 10% level.

	China	China	India	India
	(1)	(2)	(3)	(4)
	Investment	Investment	Investment	Investment
growth	0.444** (2.54)	0.413** (2.35)	0.147*** (6.30)	0.0433** (2.84)
leverage	0.945 (0.55)	0.848 (1.21)	1.983*** (3.25)	2.313*** (3.48)
size	-0.851 (-1.49)	-0.976** (-2.75)	0.0849 (0.30)	0.200 (0.82)
age	0.0739 (1.22)	0.0541*** (3.05)	0.0264 (0.60)	0.0381 (0.95)
cash flow	0.211*** (87.78)	0.155*** (6.20)	-0.0128 (-1.02)	-0.0611 (-1.66)
ppp	-0.186 (-0.49)	0.218 (0.79)	-1.413*** (-3.51)	-0.0969 (-0.08)
ppp*cash flow	-0.226*** (-22.95)	-0.150*** (-6.90)	2.469*** (4.16)	1.857* (2.02)
Tobin q_{t-1}		-0.364*** (-3.49)		-0.0224 (-0.40)
ppp*Tobin q_{t-1}		0.214** (2.77)		-0.608 (-1.16)
Tobin q_{t-1} *cash flow		0.797* (2.12)		0.0769 (1.60)
ppp*Tobin q_{t-1} *cash flow		-0.802** (-2.15)		0.193 (0.80)
Constant	5.360 (1.68)	6.505** (2.60)	-1.255 (-0.79)	-2.401 (-1.70)
R-squared	0.88	0.95	0.76	0.79
N	298	267	329	301

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table VI: China & India: Investment-Cash Flow Sensitivity - PPP firms & Non-PPP firms in 3-year Post-PPP

This table compares investment-cash flow sensitivity in between PPP firms and Non-PPP firms in 3-year Post-PPP. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Size is measured by the natural logarithm of total assets. Growth presents the changes in sales from the previous year divided by the previous year's sales. Leverage is calculated by total debt divided by total assets. Age is measured from the year of firms incorporation. PPP is a dummy variable which takes value 1 for firms invested in PPP projects, else zero for firms which are matched by industry and firm size. Tobin q_{t-1} (with one year lag) to capture investment opportunities ***, **, * indicate significant at 1%, 5%, and 10% level.

	China	China	India	India
	(1)	(2)	(3)	(4)
	Investment	Investment	Investment	Investment
growth	0.534*** (7.98)	0.452*** (9.04)	0.359** (2.54)	0.346* (1.82)
leverage	0.189 (0.64)	-0.134 (-0.65)	0.448 (1.21)	0.244 (1.64)
size	-0.0353 (-0.48)	0.0221 (0.85)	0.0805 (0.97)	0.0770 (1.27)
age	-0.00733 (-0.70)	-0.0129 (-1.38)	-0.0120 (-1.64)	-0.0153* (-1.89)
cash flow	1.514*** (6.15)	1.834*** (7.00)	-0.00456 (-0.31)	0.0107 (0.30)
ppp	0.420 (1.59)	0.622 (1.51)	-0.206** (-2.01)	-0.133 (-0.54)
ppp*cash flow	-1.145** (-2.34)	-1.872*** (-6.71)	0.540*** (9.73)	-0.0259 (-0.04)
Tobin q_{t-1}		0.148 (1.03)		0.0403* (1.87)
ppp*Tobin q_{t-1}		-0.193 (-1.34)		-0.0912 (-0.75)
Tobin q_{t-1} *cashflow		-0.266* (-1.86)		-0.0114 (-1.21)
ppp*Tobin q_{t-1} *cashflow		0.474*** (3.06)		0.382 (1.12)
Constant	-0.0414 (-0.08)	-0.467 (-1.07)	-0.396 (-0.54)	-0.329 (-0.69)
R-squared	0.76	0.84	0.42	0.55
N	100	94	122	120

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table VII: China & India: Reason for Differences in Investment-Cash Flow Sensitivity - PPP firms & Non-PPP firms in PPP years

This table uses Slope Difference Test to probe three-way interactions Tobin q_{t-1} *ppp*cash flow in table V. The purpose of this test is to show whether the differences between PPP firms and NonPPP firms in Investment-Cash flow Sensitivity are caused by underinvestment or overinvestment. PPP is a dummy variable which takes value 1 for firms invested in PPP projects, else zero for firms which are matched by industry and firm size. Tobin q_{t-1} to capture investment oportunities. Adjusted P is a Bonferroni adjusted p-value which accounts for the fact that there are six post-hoc tests. This is a conservative adjustment by multiplying each of the p-values by the number of tests. ***, **, * indicate significant at 1%, 5%, and 10% level.

Panel A					
China - Slopes	Coef.	Std. Err.	t	P value	Adjusted P
(PPP & high tobin q)-(PPP & low tobin q)	-0.023938	0.0125622	-1.91	0.077	0.462
(PPP & high tobin q)-(Non-PPP & high tobin q)	-3.429492	1.511527	-2.27	0.04	0.24
(PPP & low tobin q)-(Non-PPP & low tobin q)	0.0402542	0.1057821	0.38	0.709	4.254
(Non-PPP & high tobin q)-(Non-PPP & low tobin q)	3.445808	1.624335	2.12	0.052	0.312
(PPP & high tobin q)-(Non-PPP & low tobin q)	0.0163163	0.1134195	0.14	0.888	5.328
(PPP & low tobin q)-(Non-PPP & high tobin q)	-3.405554	1.519694	-2.24	0.042	0.252
Panel B					
India - Slopes	Coef.	Std. Err.	t	P value	Adjusted P
(PPP & high tobin q)-(PPP & low tobin q)	2.720288	2.511246	1.08	0.295	1.77
(PPP & high tobin q)-(Non-PPP & high tobin q)	3.277495	1.136362	2.88	0.011	0.066*
(PPP & low tobin q)-(Non-PPP & low tobin q)	1.331678	1.512173	0.88	0.392	2.352
(Non-PPP & high tobin q)-(Non-PPP & low tobin q)	0.7744708	0.4832949	1.6	0.129	0.774
(PPP & high tobin q)-(Non-PPP & low tobin q)	4.051966	1.307016	3.1	0.007	0.042**
(PPP & low tobin q)-(Non-PPP & high tobin q)	0.5572071	1.571775	0.35	0.728	4.368
Hypothesis Development Matrix					
China	PPP - PPP	PPP - Non-PPP			
High q - Low q	Not significant	No prediction			
High q - High q	No prediction	Not significant			
Low q - Low q	No prediction	Not significant			
India	PPP - PPP	PPP - Non-PPP			
High q - Low q	Not significant	No prediction			
High q - High q	Not prediction	(+): PPP has more underinvestment than Non-PPP			
Low q - Low q	Not prediction	Not significant			

Table VIII: China & India: Reason for Differences in Investment-Cash Flow Sensitivity - PPP firms & NonPPP firms in 3-year PostPPP

This table uses Slope Difference Test to probe three-way interactions Tobin q_{t-1} *ppp*cash flow in TableVI. The purpose of this test is to show whether the differences between PPP firms and NonPPP firms are caused by underinvestment or overinvestment. PPP is a dummy variable which takes value 1 for firms invested in PPP projects, else zero for firms which are matched by industry and firm size. Tobin q_{t-1} to capture investment opportunities. Adjusted P is a Bonferroni adjusted p-value which accounts for the fact that there are six post-hoc tests. This is a conservative adjustment by multiplying each of the p-values by the number of tests. ***,**,* indicate significant at 1%, 5%, and 10% level.

Panel A					
China - Slopes	Coef.	Std. Err.	t	P value	Adjusted P
(PPP & high tobin q)-(PPP & low tobin q)	1.196554	0.4544944	2.63	0.008	0.048**
(PPP & high tobin q)-(Non-PPP & high tobin q)	0.3768788	0.4853863	0.78	0.437	2.622
(PPP & low tobin q)-(Non-PPP & low tobin q)	-2.343105	0.4258289	-5.5	0	0***
(Non-PPP & high tobin q)-(Non-PPP & low tobin q)	-1.523429	0.8196261	-1.86	0.063	0.378
(PPP & high tobin q)-(Non-PPP & low tobin q)	-1.146551	0.5426941	-2.11	0.035	0.21
(PPP & low tobin q)-(Non-PPP & high tobin q)	-0.8196756	0.4606585	-1.78	0.075	0.45
Panel B					
India - Slopes	Coef.	Std. Err.	t	P value	Adjusted P
(PPP & high tobin q)-(PPP & low tobin q)	1.436873	1.335136	1.08	0.282	1.692
(PPP & high tobin q)-(Non-PPP & high tobin q)	1.349488	0.5386746	2.51	0.012	0.072*
(PPP & low tobin q)-(Non-PPP & Low tobin q)	-0.131372	0.7807278	-0.17	0.866	5.196
(Non-PPP & high tobin q)-(Non-PPP & Low tobin q)	-0.0439863	0.0362553	-1.21	0.225	1.35
(PPP & high tobin q)-(Non-PPP & low tobin q)	1.305501	0.5565877	2.35	0.019	0.114
(PPP & low tobin q)-(Non-PPP & high tobin q)	-0.0873857	0.7968079	-0.11	0.913	5.478
Hypothesis Matrix					
China	PPP - PPP	PPP - Non-PPP			
High q - Low q	(+): PPP has underinvestment	No prediction			
High q - High q	No prediction	Not significant			
Low q - Low q	No prediction	(-): PPP has less overinvestment than Non-PPP			
India	PPP - PPP	PPP - Non-PPP			
High q - Low q	Not significant	No prediction			
High q - High q	No prediction	(+): PPP has more underinvestment than Non-PPP			
Low q - Low q	No prediction	Not significant			

Table IX: The Effect of Government on Investment-Cash Flow Sensitivity in the 5-year post-PPP period

This table presents the effects of Government on investment-cash flow sensitivity on PPP firms in the 5-year post-PPP period. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow(CF) denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Size is measured by the natural logarithm of total assets. Growth presents the changes in sales from the previous year divided by the previous year's sales. Leverage is calculated by total debt divided by total assets. Age is measured from the year of firms incorporation. Public Uncertainty(PU) is a dummy variable which takes value 1 for PPP firms in general election years, else zero for PPP firms in non-election years. Public Equity (PE), which is measured by the proportion of total PPP investment belongs to the government, is to capture Government Involvement in PPP projects. ***, **, * indicate significant at 1%, 5%, and 10% level.

	China	China	India	India
Investment	(1)	(2)	(3)	(4)
cash flow	5.156445*** 9.36	0.1289613*** 4.03	0.3408282 0.74	0.0752702 0.35
leverage	0.687507 0.76	0.7961698 1.32	0.1243232 0.44	0.0131883 0.05
size	0.0354336 0.3	0.3224642*** 4.11	0.1358801*** 7.27	0.1460549*** 3.62
age	-0.0155917 -1.21	-0.0330066*** -6.49	-0.0214839*** -3.05	-0.0313892** -2.46
Tobin q_{t-1}	0.002764 0.42	0.0058594** 2.68	-0.1298559** -2.83	-0.1004318** -2.28
Public Equity(PE)	3.413981*** 9.52		-0.5232988*** -3.64	
PE*CF	-12.64425*** -8.93		3.015986*** 8.81	
Public Uncertainty (PU)		-0.55956* -1.89		-0.9270178*** -3.3
PU*CF		4.911158*** 4.68		2.667728*** 4.64
Constant	-1.355947 -1.51	-2.087113*** -2.91	-0.5964175* -1.8	-0.4898835 -1.4
R-squared	0.6211	0.6905	0.2653	0.4272
N	50	50	52	52

Table X: China & India: Reasons for the Effects of Public Equity on Investment-Cash Flow Sensitivity in 5-year Post PPP

This table uses Slope Difference Test to probe three-way interactions q_dummy*publicequity*cashflow and insider ownership*publicequity*cashflow. The purpose of this test is to show whether Government involvement in PPP projects is more likely to overinvest or underinvest. Public Equity, which is measured by the proportion of total PPP investment belongs to the government, is to capture Government Involvement in PPP projects. The adjusted-P is the Bonferroni adjusted p-value which accounts for the fact that there are six post-hoc tests. This is a conservative adjustment by multiplying each of the p-values by the number of tests. ***, **, * indicate significant at 1%, 5%, and 10% level.

Panel A: China - Slopes		Coef.	Std. Err.	t	P value	Adjusted P
(high q & high PE)-(high q & low PE)		1.40439	1.239809	1.13	0.257	1.542
(high q & high PE)-(low q & high PE)		0.9158596	1.383087	0.66	0.508	3.048
(high q & low PE)-(low q & low PE)		-4.637005	1.452782	-3.19	0.001	0.006***
(low q & high PE)-(low q & low PE)		-4.148475	1.523293	-2.72	0.006	0.036**
(high q & high PE)-(low q & low PE)		-3.232615	1.178679	-2.74	0.006	0.036**
(high q & low PE)-(low q&high PE)		-0.4885306	0.1545007	-3.16	0.002	0.012**
Panel B: India - Slopes		Coef.	Std. Err.	t	P value	Adjusted P
(high q & high PE)-(high q & low PE)		0.8105477	0.3934108	2.06	0.039	0.234
(high q & high PE)-(low q & high PE)		-1.126589	0.6046948	-1.86	0.062	0.372
(high q & low PE)-(low q & low PE)		-1.126589	0.6046948	-1.86	0.062	0.372
(low q & high PE)-(low q & low PE)		0.8105477	0.3934108	2.06	0.039	0.234
(high q & high PE)-(low q & low PE)		-0.3160415	0.9928533	-0.32	0.75	4.5
(high q & low PE)-(low q & high PE)		-1.937137	0.2347298	-8.25	0***	0

Hypothesis matrix			
China	High PE - High PE	High PE - Low PE	Low PE - Low PE
High q - Low q	Not significant	No prediction	(-): Low PE has overinvestment
High q - High q	No prediction	Not significant	No prediction
Low q - Low q	No prediction	(-): High PE has less overinvestment	No prediction
India	High PE - High PE	High PE - Low PE	Low PE - Low PE
High q - Low q	Not significant	No prediction	Not significant
High q - High q	No prediction	Not significant	No prediction
Low q - Low q	No prediction	Not significant	No prediction

Table XI: The Effects of Contract Mechanism on Investment-Cash Flow Sensitivity in PPP firms in 5 year post PPP

This table presents the effects of PPP contract mechanism on investment-cash flow sensitivity. Investment is measured by the changes in gross fixed assets, divided by the previous years' fixed assets. Cash flow (CF) denotes income before extraordinary items, depreciation and amortization, divided by the previous years' fixed assets. Firm size is measured by the natural logarithm of total assets. $tobin\ q_{t-1}$ (with one year lag) is to capture investment opportunities. Leverage is calculated by total debt divided by total assets. Type PPI is a dummy variable which takes value 1 for PPP firms with Greenfields projects and takes value zero for those with Concession projects. Awarding method is a dummy variable which takes value 1 for PPP firms with their projects awarded by direct negotiation and takes value zero for those with their projects awarded by competitive biddings. Revenue sources is a dummy variable which takes value 1 for PPP firms having revenue from fixed payments and takes value zero for those having revenue from user fees. ***, **, * indicate significant at 1%, 5% and 10% level.

Investment	China	China	China	India	India	India
	(1)	(2)	(3)	(4)	(5)	(6)
cash flow	0.1062571	4.968007***	0.112517	1.397363	-0.2453733	-0.3703556
	1.58	7.86	1.67	0.72	-0.26	-0.67
leverage	3.250357***	2.744749***	2.489975***	0.1540991	-0.1362653	-0.0383351
	3.04	3.22	2.78	0.24	-0.11	-0.07
size	-0.3416581*	-0.0010387	-0.2798495	0.5329437	1.734775*	-0.0333713
	-1.74	0	-1.32	1.52	1.75	-0.09
age	0.0084663	-0.0216923	0.0017489	-0.033012	-0.1180828	-0.0580257**
	0.5	-0.96	0.11	-1.36	-1.51	-2.36
$tobin\ q_{t-1}$	0.0185438	-0.2133869*	0.0174922	-0.1041486*	-0.0766985	-0.1041234
	1.02	-1.75	0.95	-1.72	-0.82	-1.64
type ppi==Greenfields	-0.1511441			-0.0531453		
	-0.34			-0.06		
Greenfields*CF	3.011868***			-0.4970351		
	6.2			-0.26		
awardingmethod==Direct negotiation		0.6870778			1.508031	
		1.61			1.3	
Negotiation*CF		-4.748544***			1.176363	
		-7.6			1.1	
revenue sources==User fees			-1.355718***			-1.86917*
			-3.19			-2.00
User fees*CF			3.078586***			3.930056***
			6.26			3.45
Constant	1.398616	-0.0560553	1.825835	-3.570801	-15.39362*	0.9719648
	1.02	-0.03	1.26	-1.52	-1.79	0.35
N	50	45	50	52	40	52

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